

Priya Musale

priyamusale04@gmail.com | (662) 694-1422 | Bothell, WA | linkedin.com/in/priyamusale

SUMMARY

MS CS researcher at UW Bothell working on LLM personalization and clustering for mental health AI with over 5 years of building scalable data and software pipelines.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, MATLAB, Perl

Data Engineering: Databricks, PySpark, AWS Glue, Pandas, NumPy

Machine Learning: NLP, RAG, LLM Fine-tuning, Meta LLaMA, Hugging Face, PyTorch

Databases: Redshift, PostgreSQL, Oracle, Teradata, KDB (time-series)

Tools & Infrastructure: Docker, Kubernetes, Kafka, Zookeeper, SSH, GitHub

RESEARCH & PUBLICATIONS

- Submitted an IEEE conference paper on Hierarchical Deep Embedded Clustering (HDEC), a pipeline combining SBERT embeddings, autoencoders, and KL divergence refinement, applied to mental health conversational AI.
- Discovered novel behavioral phenotypes through unsupervised clustering, including a previously unidentified Caregiver Burnout cluster.
- Fine-tuned Meta LLaMA to evaluate therapeutic conversation summaries for empathy and clinical accuracy, building a monitoring framework for mental health support tools.

EDUCATION

M.S. Computer Science and Software Engineering | University of Washington, Bothell

Sep 2025 - Present

GPA: 3.98

- Researching temporal trajectory modeling for mental health conversational AI in Dr. Dong Si's iCare DAIS Lab.

B.S. Computer Science and Engineering | Mississippi State University

- Student researcher at NSPARC, researching IT software and presenting findings company-wide.
- Resident Advisor, fostering a supportive campus community while building leadership and conflict resolution skills.

WORK EXPERIENCE

Data Engineer | Nationwide

Phoenix, AZ (Remote) | June 2023 - August 2025

- Migrated 12 critical AWS Glue PySpark jobs to Databricks, cutting batch processing runtime from 8 hours to 2 hours and enabling faster downstream data availability for analytics and modeling.
- Slashed failure tickets from 1,000+ to <10 per month and eliminated on-call escalations by optimizing 150+ Kubernetes and Docker batch jobs.
- Accelerated the onboarding of new client data sources by building Python automation to replace manual pipeline triggers.
- Drove on-time delivery across 5 concurrent projects by aligning a 24-member onshore and offshore team with stakeholder priorities, hitting 100% of milestones.

Full Stack Software Engineer | LinkedIn

Sunnyvale, CA (Remote) | May 2022 - June 2023

- Upgraded core backend architecture (Python 3.7 to 3.10) for the HireIn platform, resolving critical vulnerabilities and improving latency for features used by millions of recruiters.
- Revamped 50+ production features in the HireIn platform, streamlining recruiter and candidate workflows and directly strengthening LinkedIn's core hiring product used by millions.
- Improved a high-traffic internal platform by overhauling JavaScript queries and frontend logic, trimming page load time from ~4 seconds to ~3 seconds, and reducing recruiter friction.

Python Developer | JPMorgan Chase and Co.

Houston, TX | February 2021 - May 2022

- Architected automated data quality reporting across 500,000+ records, surfacing datasets an hour earlier per run and empowering traders to make faster, better-informed decisions.
- Designed and deployed Python APIs to retrieve KDB time-series data, compute profit and loss metrics, and expose results as JSON objects, powering business stock decision-making tools.
- Optimized Python and SQL jobs across multiple automated workflows, shaving 30+ minutes per run and improving analyst access to time-sensitive datasets.